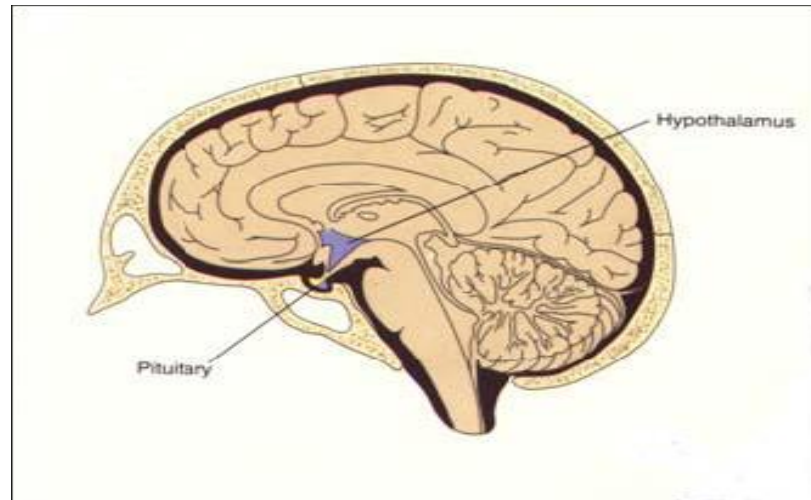


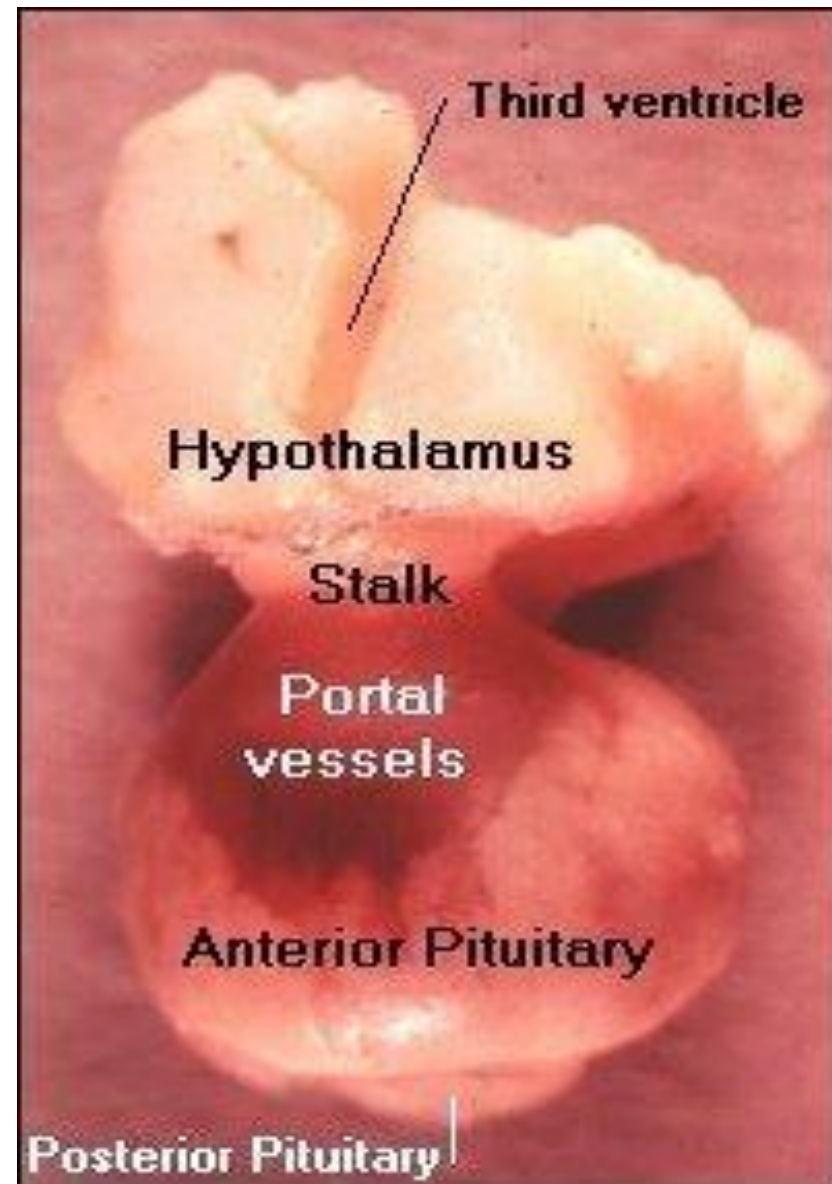
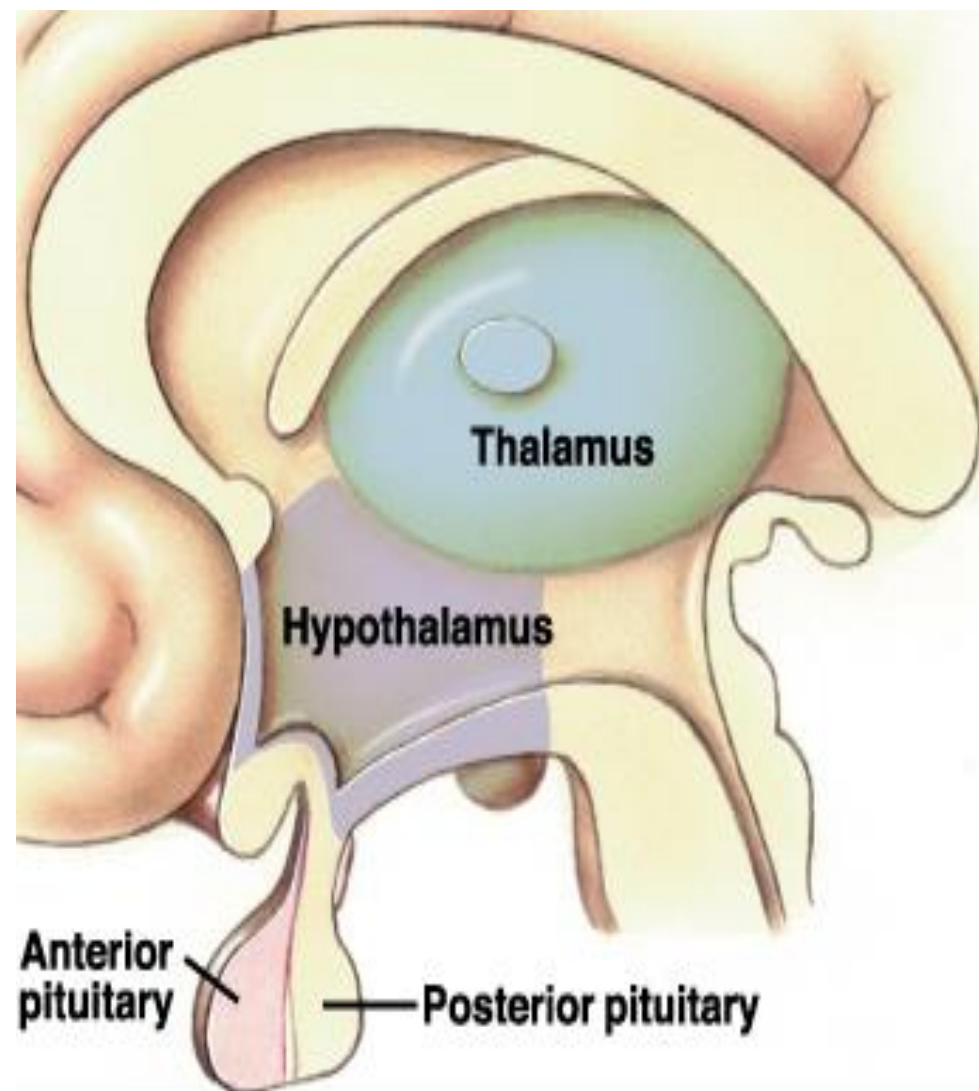
Physiology of the Endocrine System



Hypothalamus and Pituitary Gland

Hypothalamus

- Part of the diencephalon (Component of the limbic system)
- Forms the floor and the lateral wall of the 3rd ventricle
- Located below the thalamus
- BBB in this area is weaker than the rest of the brain
- Plays important role in controlling homeostasis



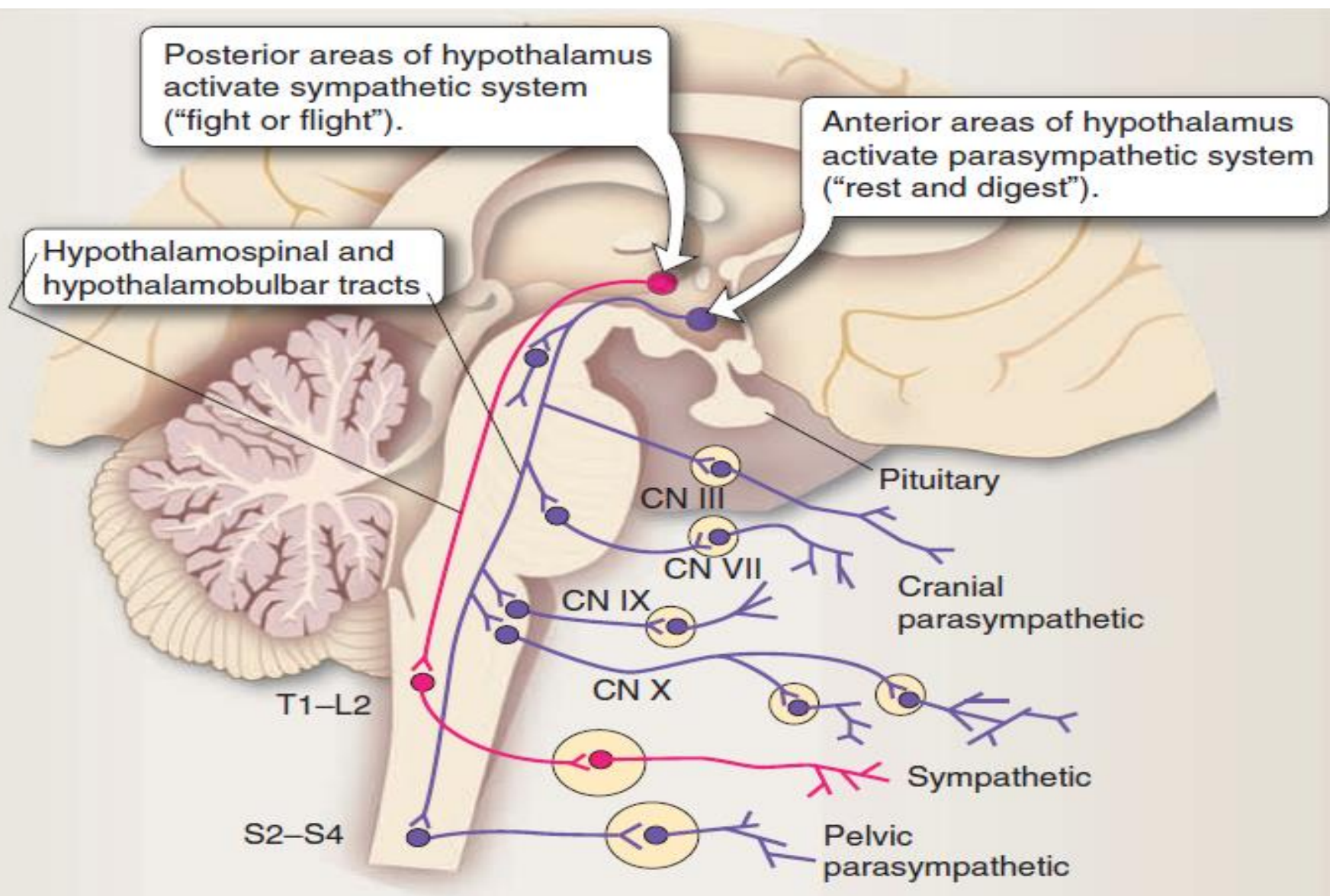
Hypothalamus: Functions

1. Endocrine control

- Controls Adenohypophyseal secretions
- Controls Neurohypophyseal secretions

2. Autonomic control

- Anterior Nuclei → acts as a parasympathetic center
- Posterior Nuclei → acts as a sympathetic center
 - Hypothalamus receives signals from almost all parts of the body and acts as a collecting center for internal body information



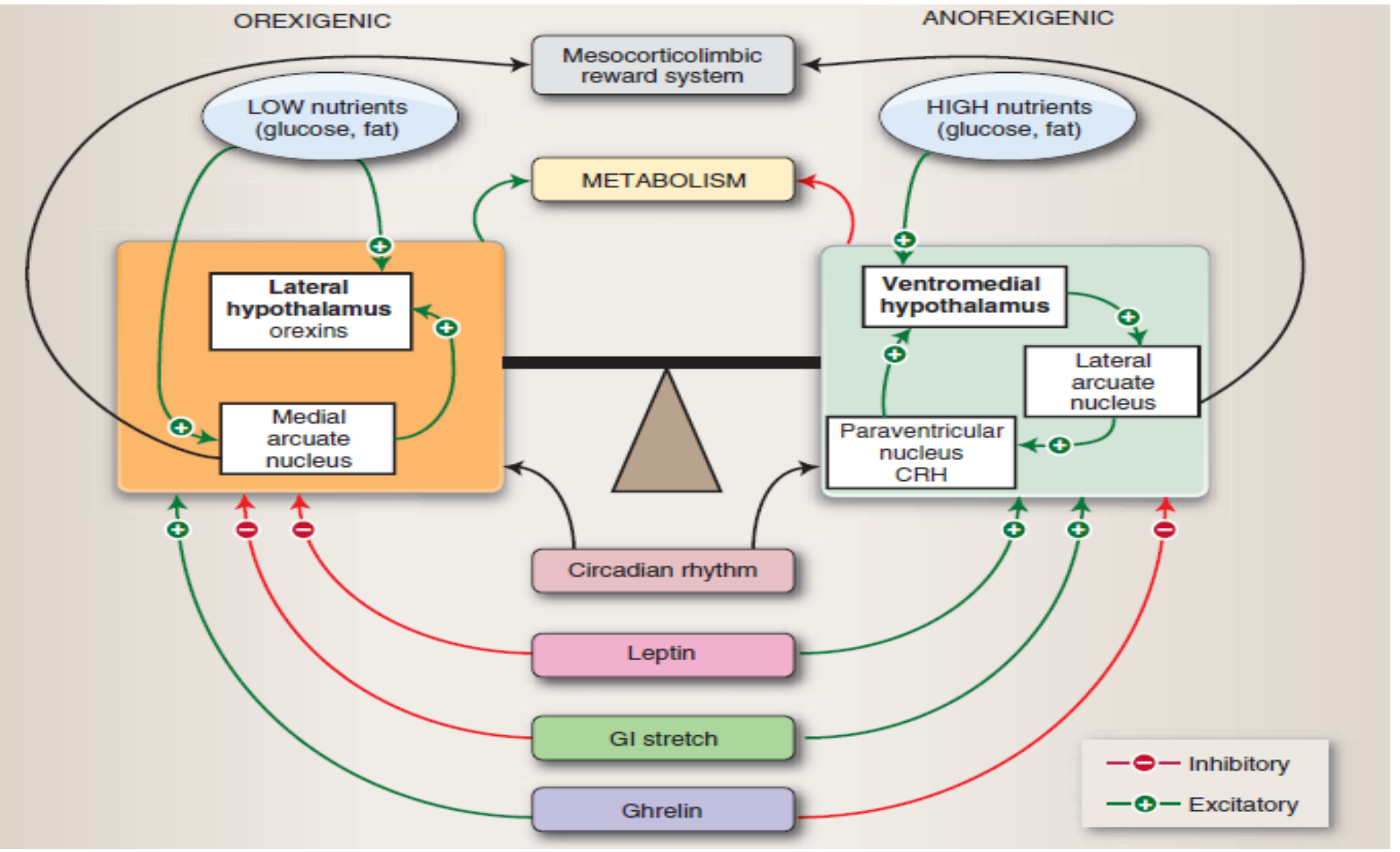
Regulation of autonomic/visceral function

Hypothalamus: Functions

3. Regulate food intake

- Lateral HT → Hunger/Feeding center
- Medial/Ventromedial HT → Satiety center

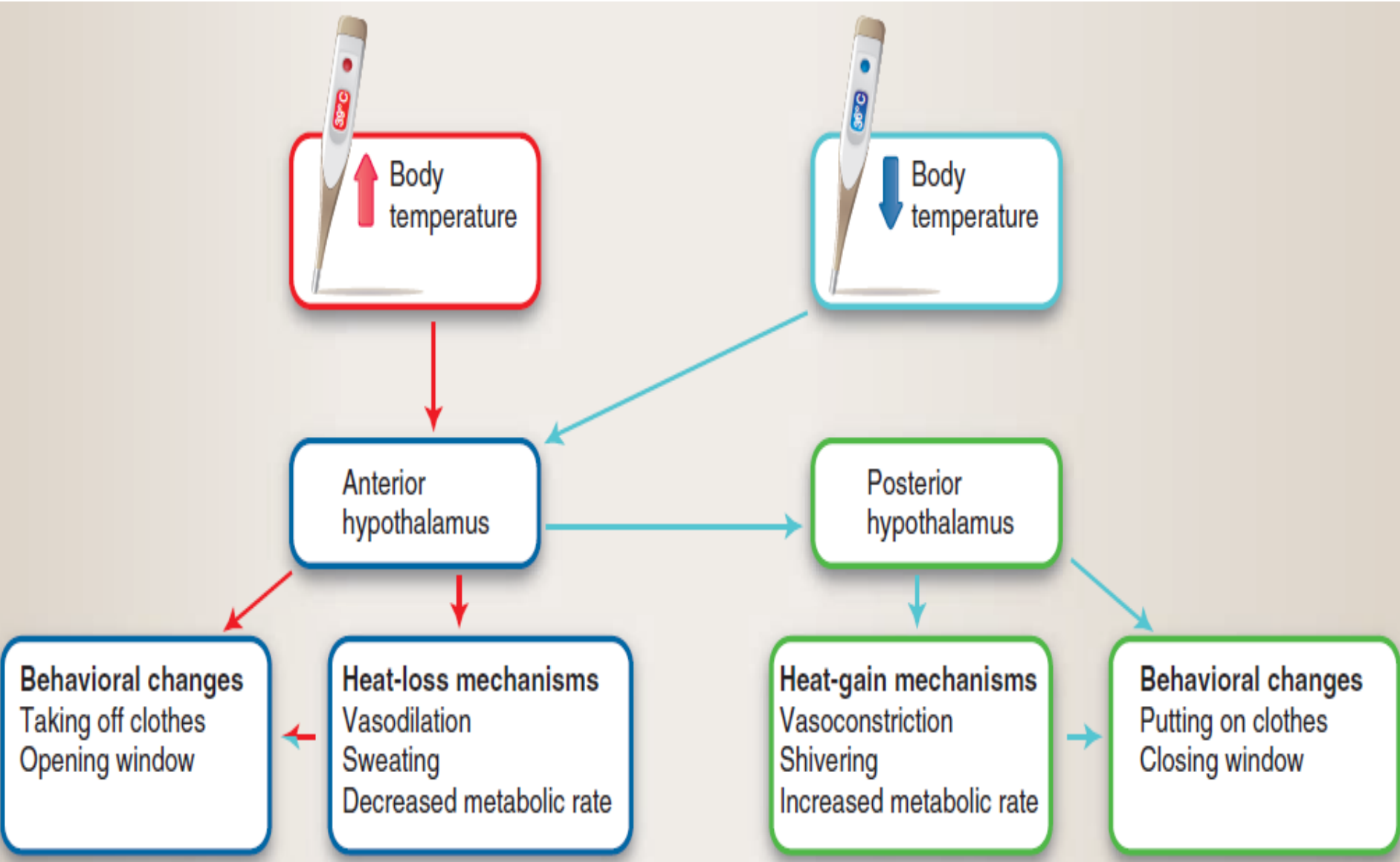
The regulation of eating and satiety by the hypothalamus



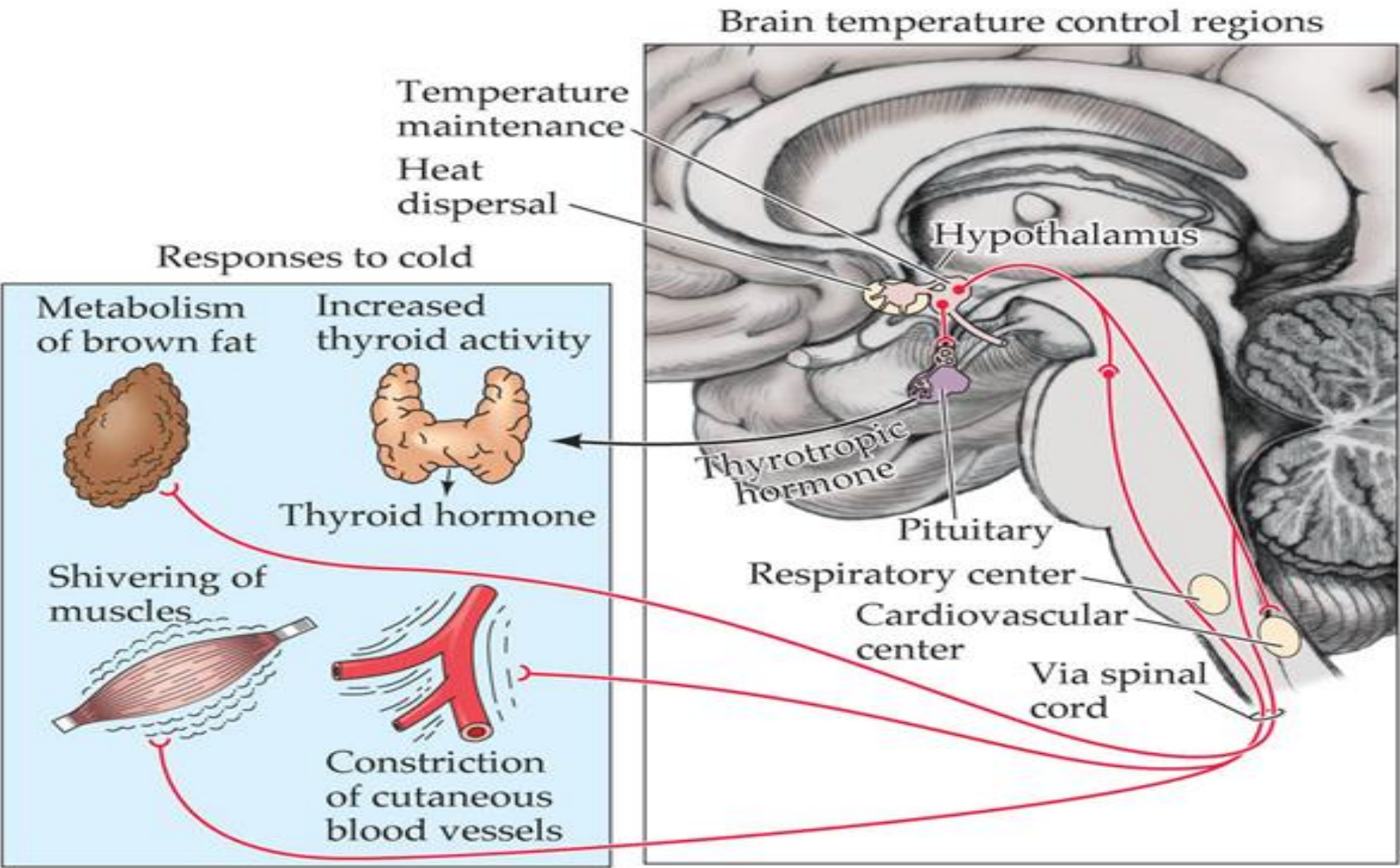
Hypothalamus: Functions

4. Regulation of body temperature

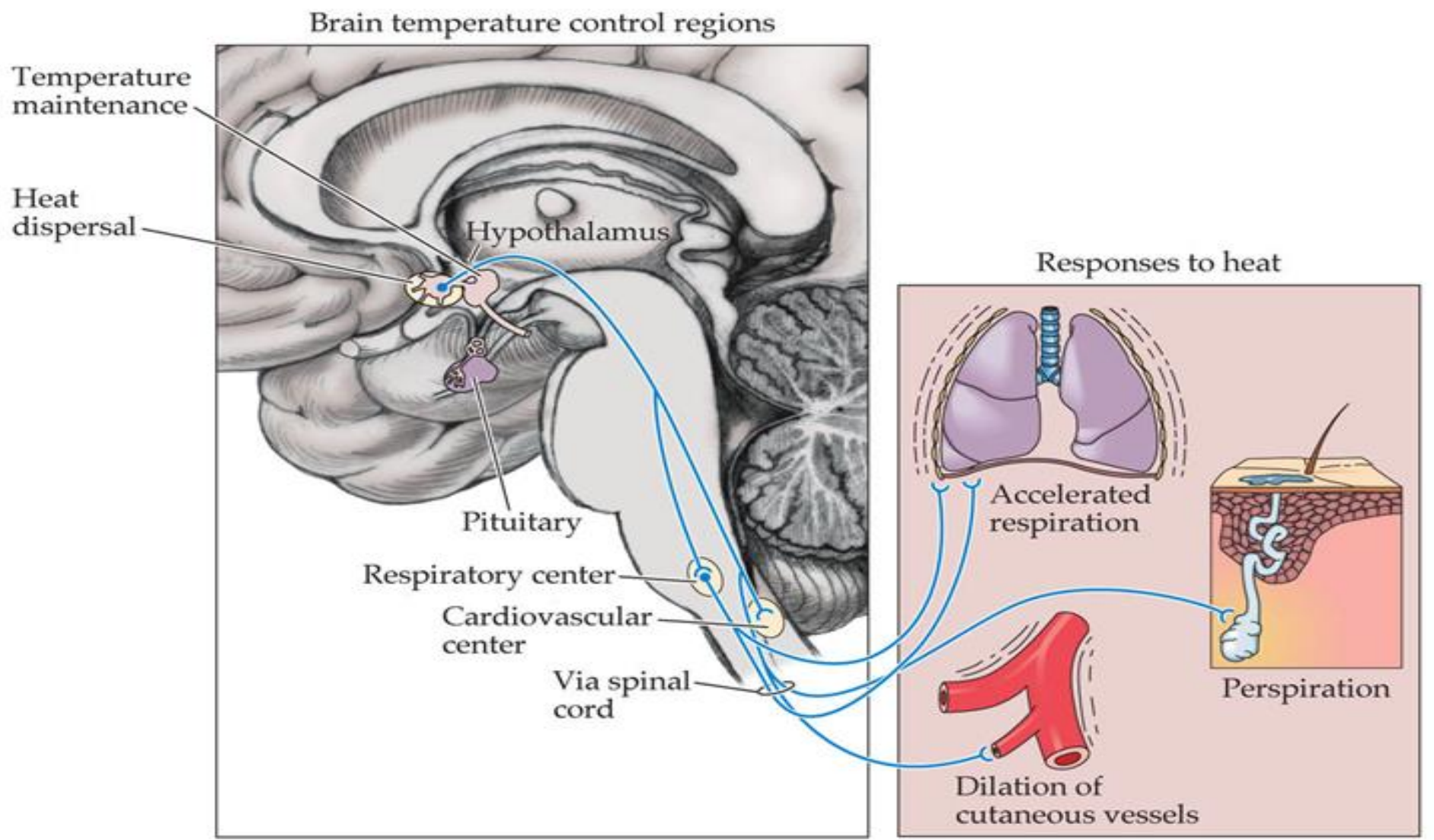
- Anterior Preoptic Area → Thermostat center
 - Anterior HT (Preoptic area) → Heat loss (dissipation) center
 - Posterior HT → Heat gain (conservation) center
 - Reciprocal connection...
- Controls secretion of hormones that influence metabolic rate



Thermoregulation systems...



Thermoregulation systems...



Hypothalamus: Functions

5. Control of Blood pressure & electrolyte composition:

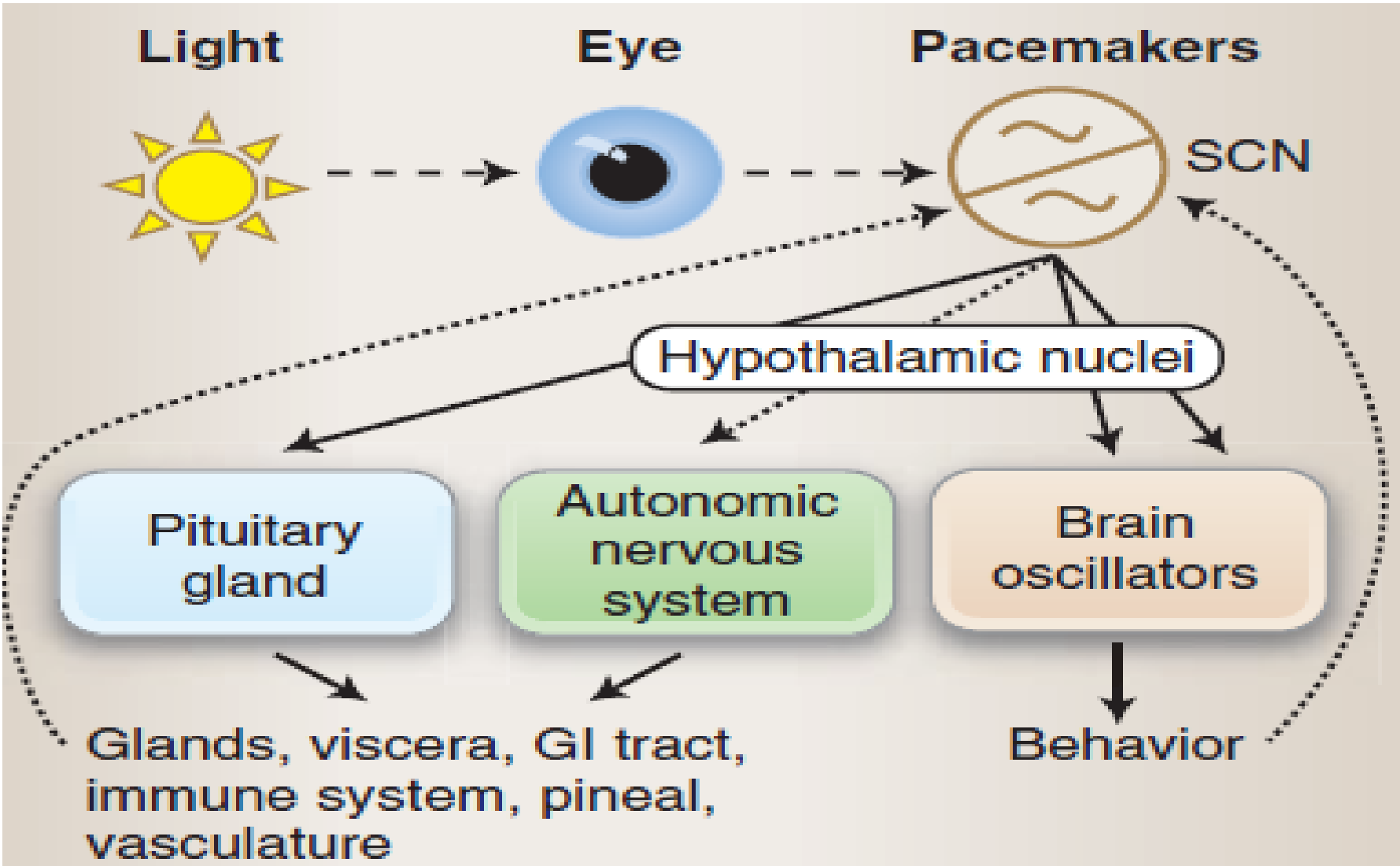
- Anterior HT → Salt appetite center (Osmoreceptors)
- Lateral HT → Thirst center
- Receive inputs from different osmoreceptors of the body
- Release of vasopressin (ADH)...
- Adjusting autonomic vasomotor tone...

Hypothalamus: Functions

6. Control of circadian rhythm and the sleep-wake cycles

- Retinohypothalamic tract → Suprachiasmatic nucleus (SCN) → Circadian clock
- Sleep →? Posterior hypothalamus
 - Hypothalamic neurons project on to reticular activating system (RAS) where sleep center is located

The regulation of the circadian rhythm by the hypothalamus



Hypothalamus: Functions

7. Control of motor responses to emotions:

- Part of the Papez circuit → Emotional expression
 - Fear + Rage + Placidity...

8. Role in learning and memory

- Part of the Papez circuit → Emotional learning and memory
- Lateral and ventromedial center → Reward center
- Periventricular center → Punishment center

9. Control of milk production and uterine contraction

- Prolactin + Oxytocin

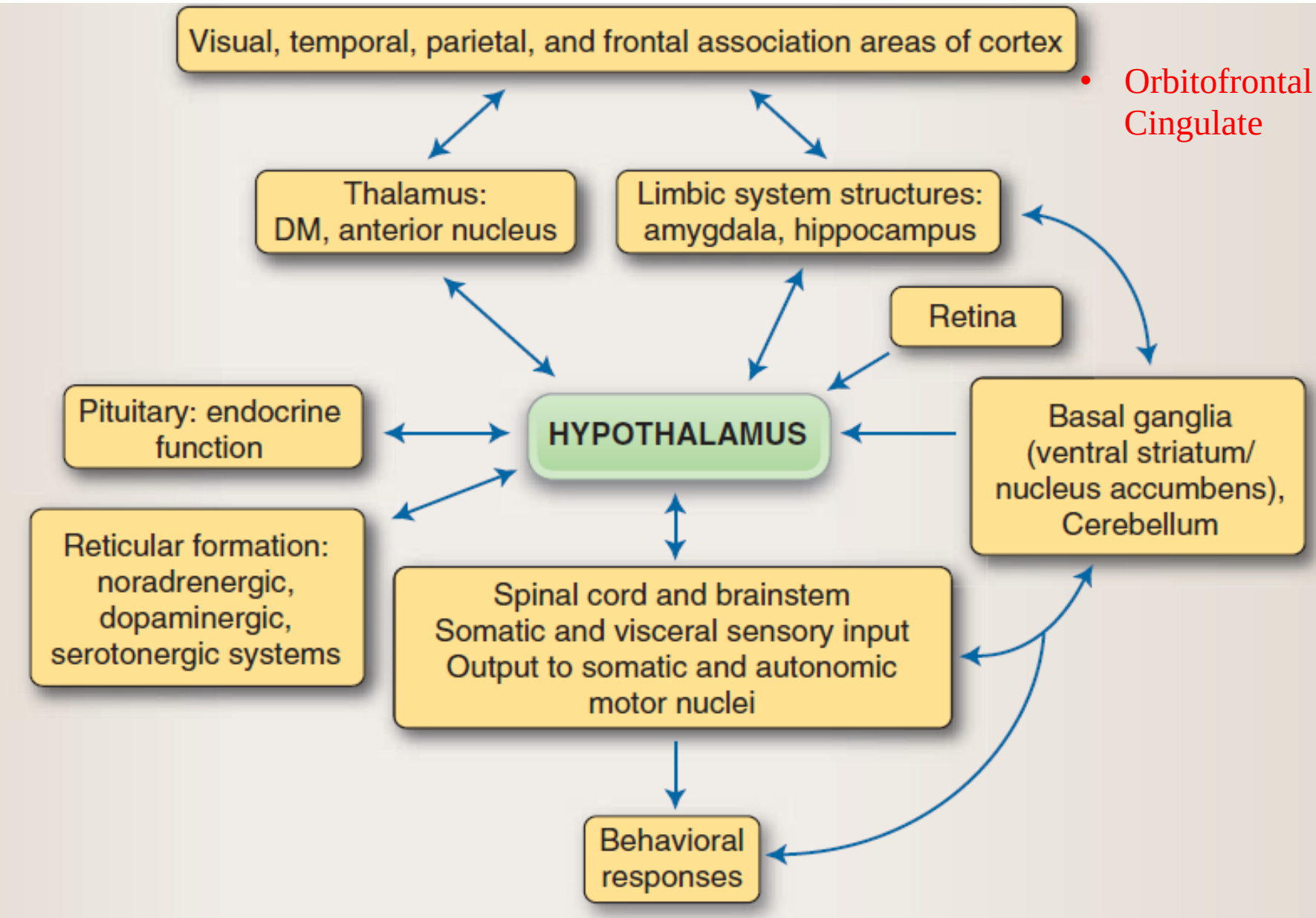
Hypothalamus: Functions

10. Control of sexual behavior + Reproduction

- Cerebral cortex → libido and sexual activity
- Limbic system → emotions associated with libido
- Hypothalamus → Arcuate nucleus → GnRH → FSH+ LH

11. Control of energy metabolism

- Regulates hunger & feeding behavior
- Regulate Autonomic control of digestion
- Regulate release of hormones → Glucocorticoids, GH, TSH...

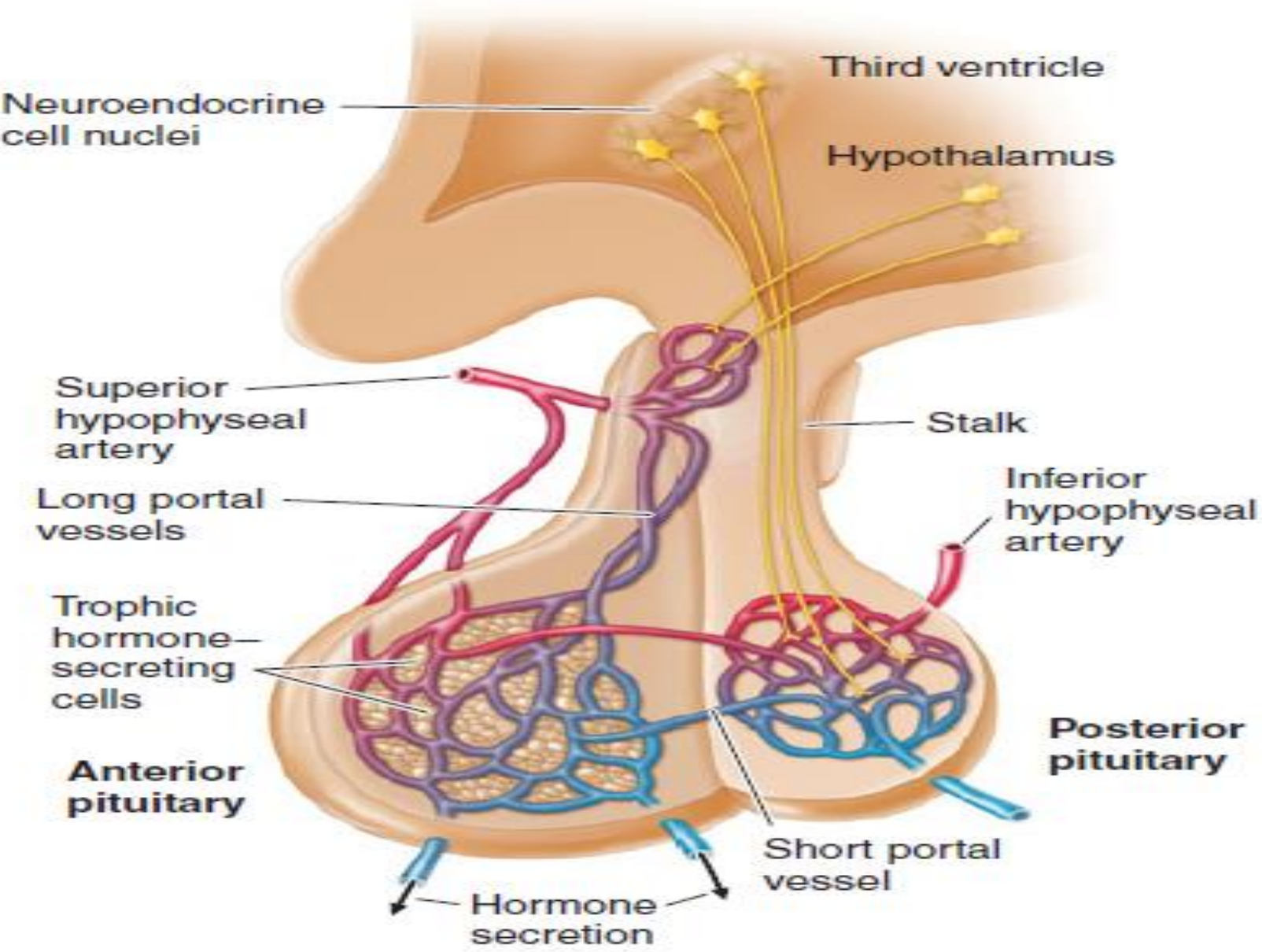


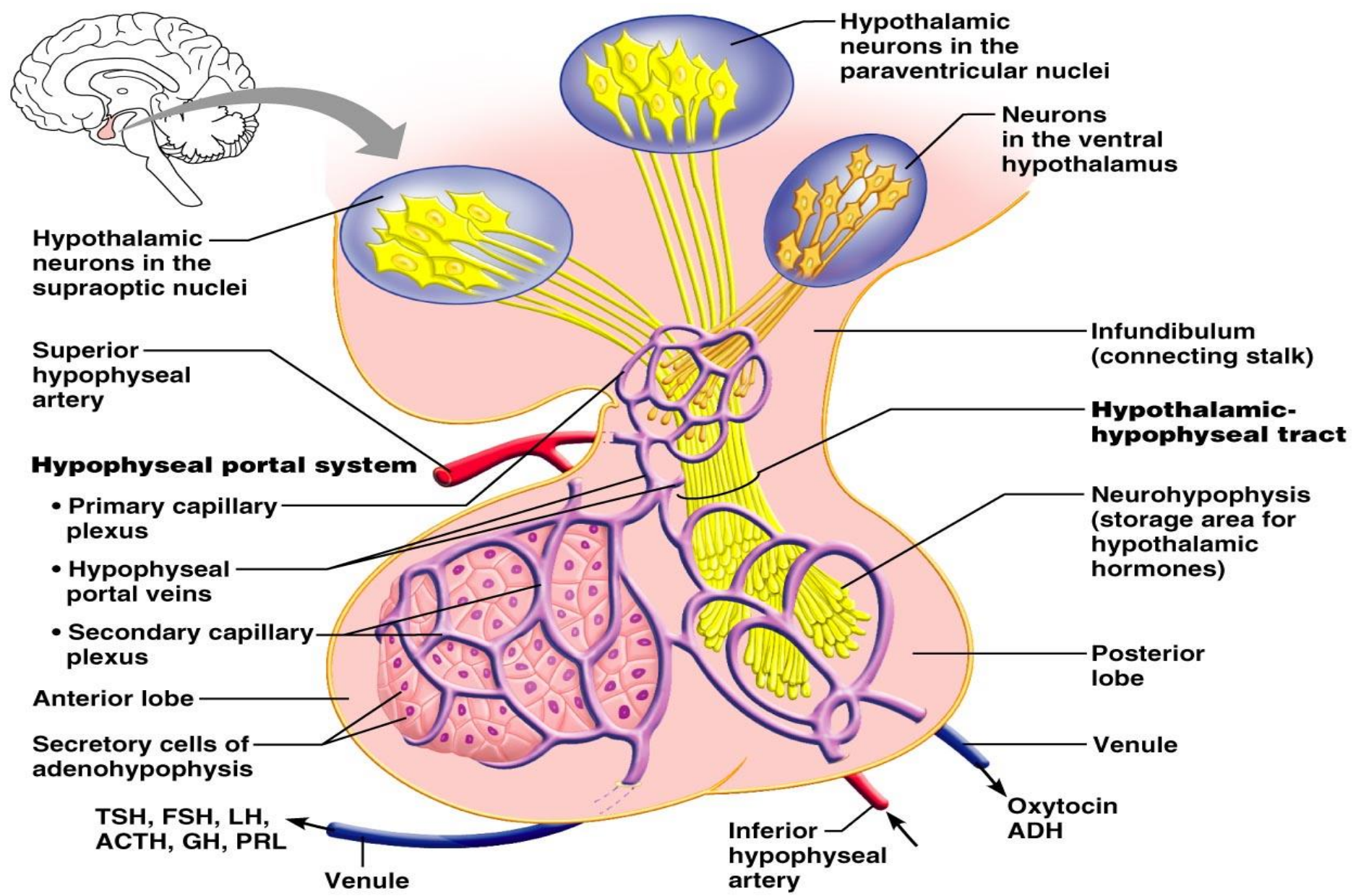
Hypothalamus

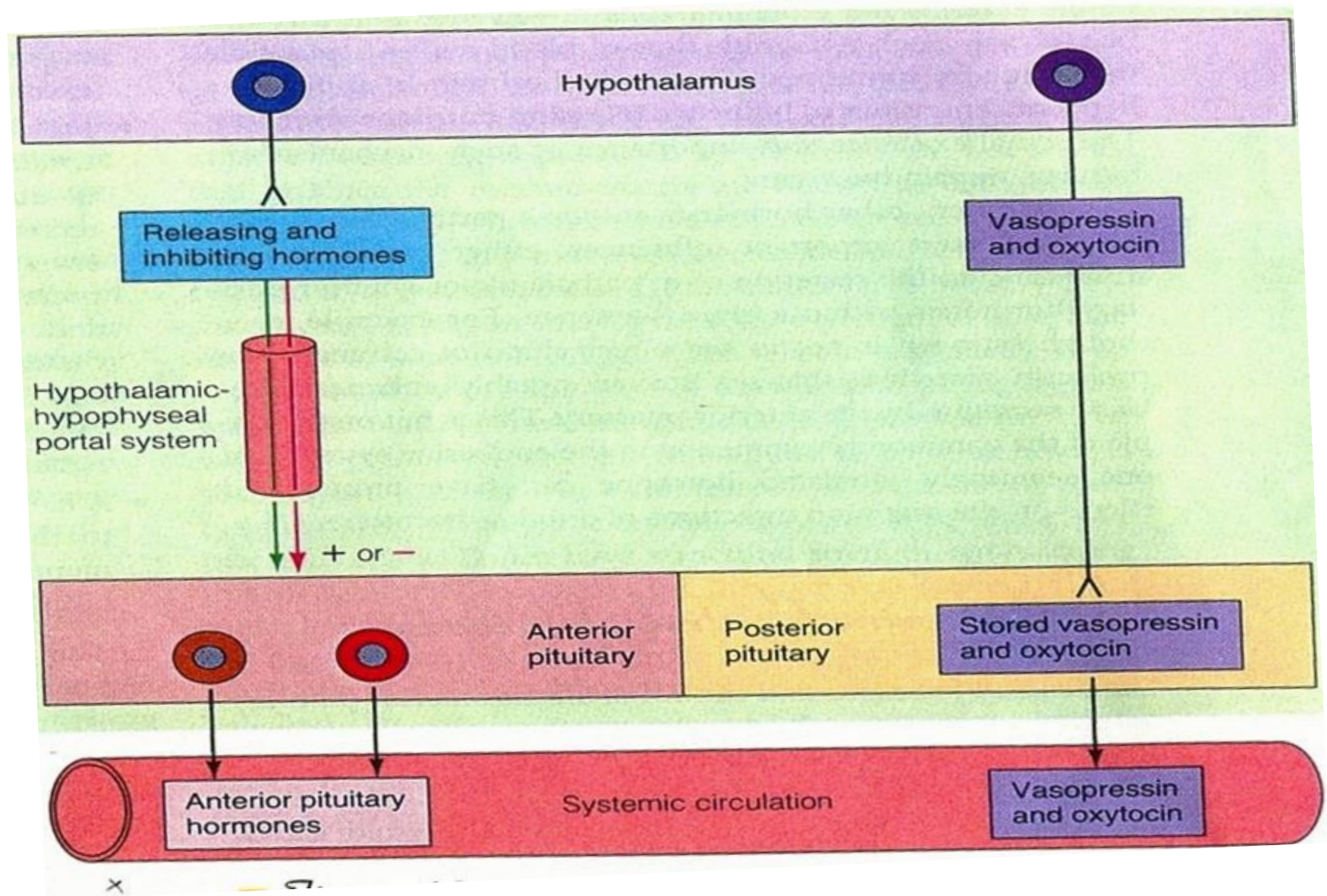
- **Endocrine Function:**
 - Neuroendocrine cells of the hypothalamus produce releasing/inhibiting hormones
 - These hormones act on the pituitary gland and
 - Induce signal transduction via second messengers
 - Stimulate synthesis and release of pituitary hormones
 - Hypothalamus is connected with the pituitary gland in two ways:
 - **Neural connection** → Hypothalamic-Hypophyseal tract → HT-PP
 - **Vascular connection** → Hypophyseal portal system → HT-AP

Hypothalamus

- Hypothalamo-Hypophyseal portal system:
 - Hypothalamic control of the anterior pituitary is achieved through hormonal regulation
 - Releasing and inhibiting hormones are transported to axon endings in the basal portion of the hypothalamus (median eminence)
 - Median eminence contains capillaries that are drained by venules to a second capillary bed in the anterior pituitary
 - Then, these hormones regulate secretions of the anterior pituitary

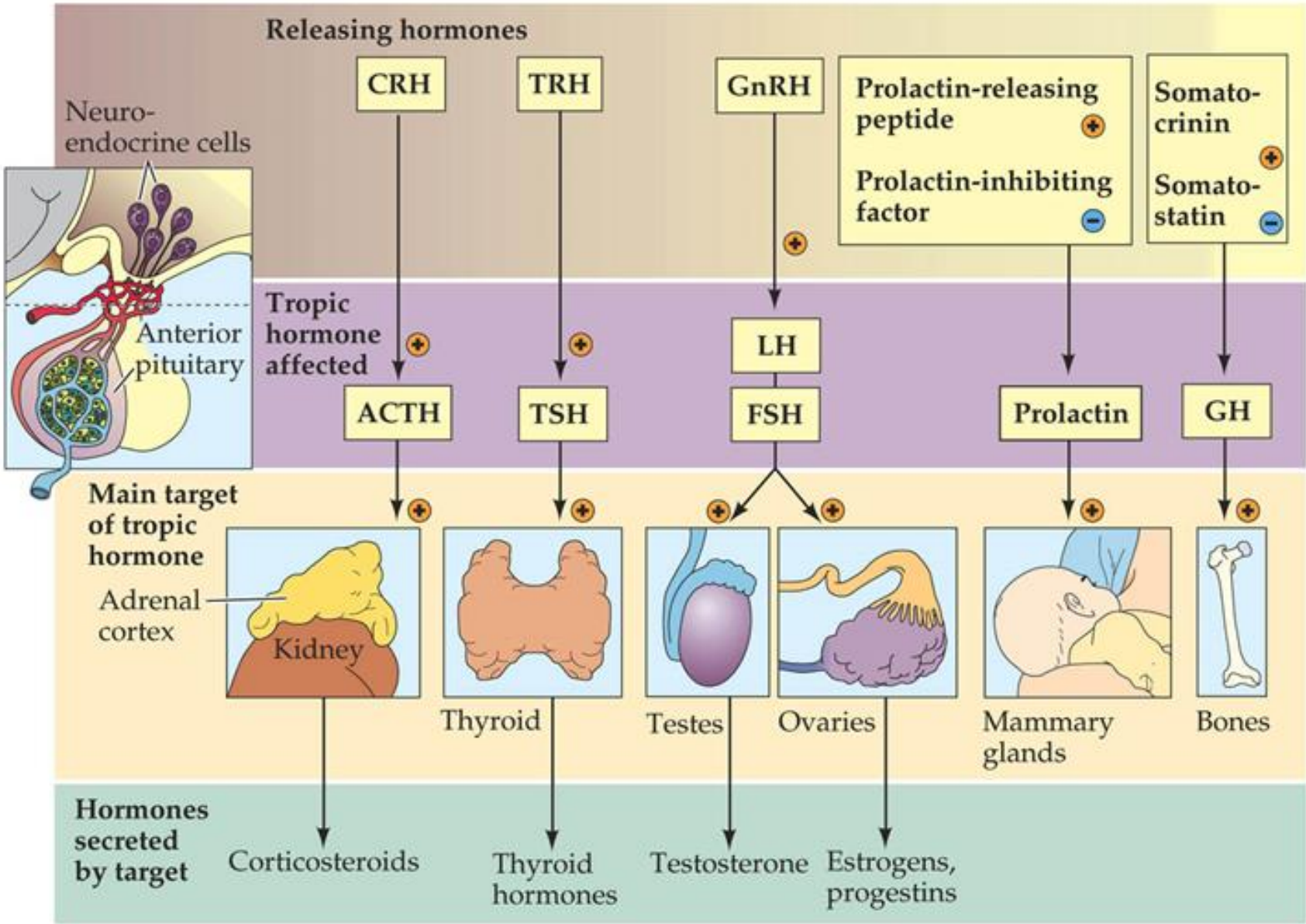






Hypothalamic releasing/inhibiting hormones

Hypothalamic hormones	Effect on the Pituitary
Corticotropin releasing hormone (CRH)	Stimulates ACTH secretion
Thyrotropin releasing hormone (TRH)	Stimulates TSH secretion
Growth hormone releasing hormone (GHRH)	Stimulates GH secretion
Somatostatin (GHRIH)	Inhibits GH (and other hormone) secretion
Gonadotropin releasing hormone (GnRH) a.k.a LHRH	Stimulates LH and FSH secretion
Prolactin releasing hormone (PRH)	Stimulates PRL secretion
Prolactin inhibiting hormone (dopamine)	Inhibits PRL secretion

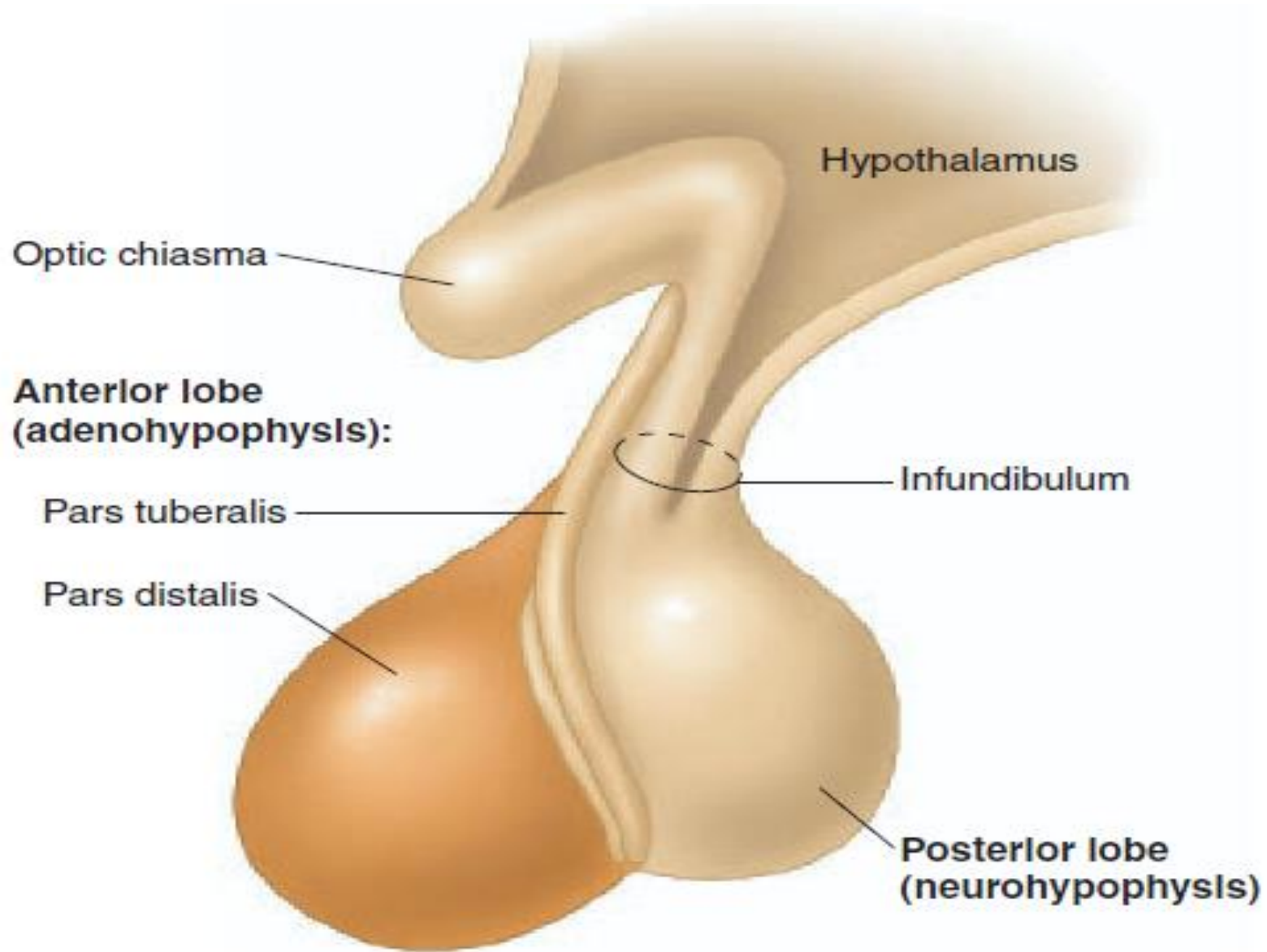


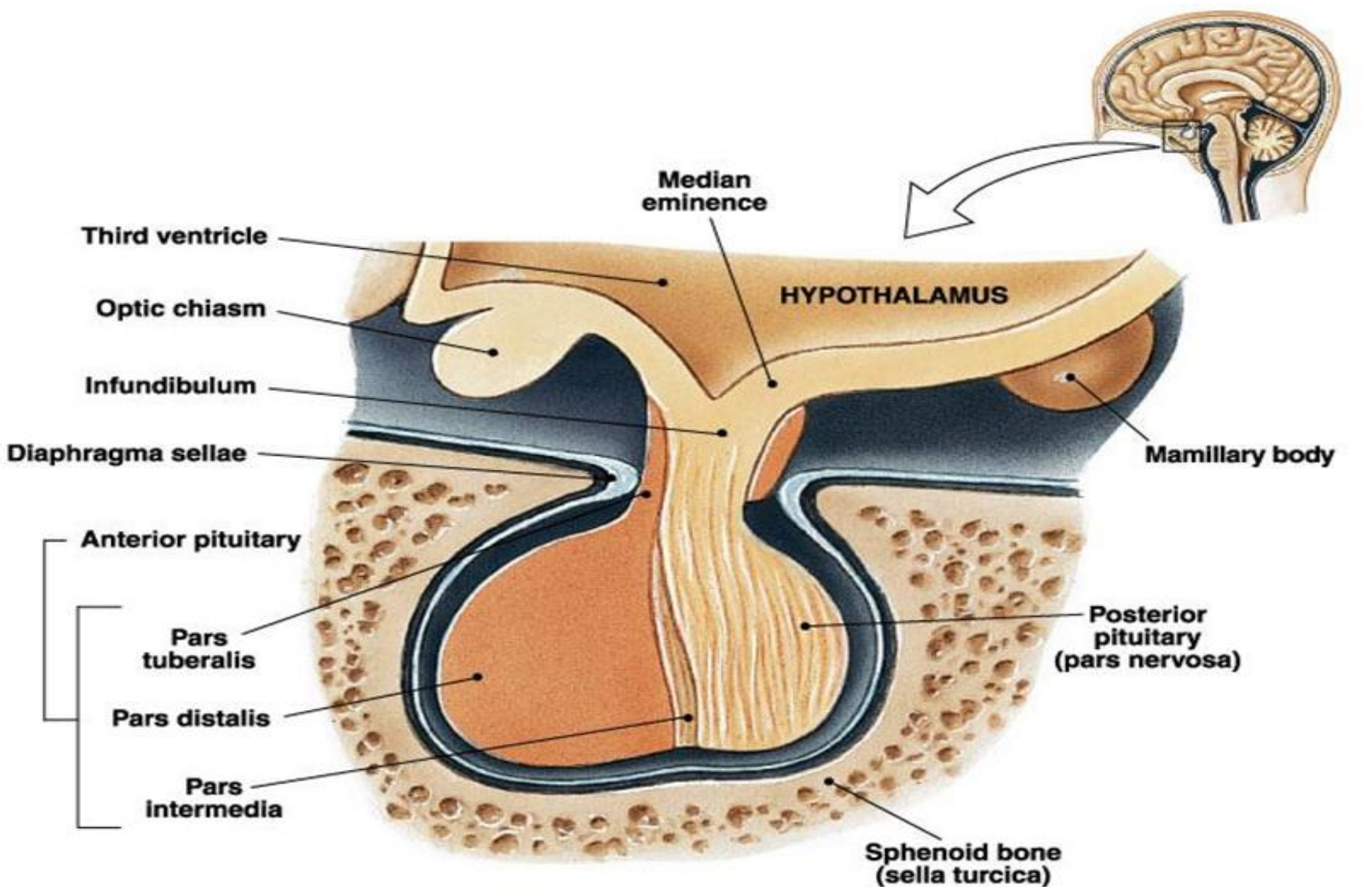
Effect of lesions in the hypothalamus

- Endocrinal disturbances
 - Diabetes Insipidus...
- Hyperthermia or hypothermia
- Hyperphagia or anorexia
- Sleep disturbances (insomnia or hypersomnia)
- Disturbances in memory and learning ability
 - Interruption of the Papez circuit
- Autonomic disturbances
- Emotional disturbances

Pituitary Gland

- Located within the sphenoid bone (sella tursica), at the base of the brain, just below the hypothalamus
- Connected to the hypothalamus by the infundibulum (Pituitary stalk) containing nerve fibers and blood vessels
- Roughly the size of a pea (diameter $\approx 1.3\text{cm}$)
- Composed of two lobes:
 1. Anterior pituitary (Adenohypophysis)
 2. Posterior pituitary (Neurohypophysis)

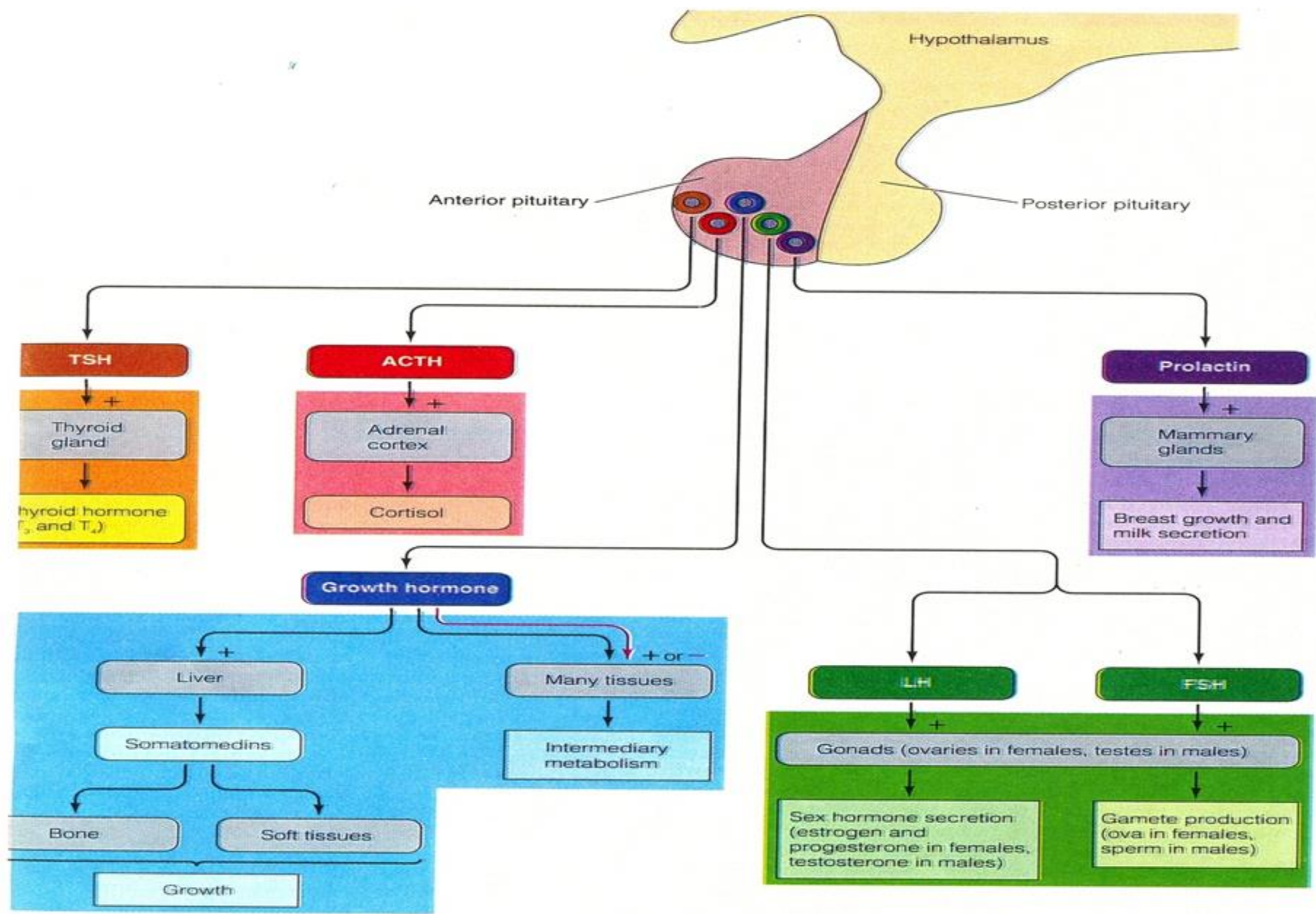




Pituitary Gland

- Anterior Pituitary
 - Consist of agranular (non-secretary) and granular (secretary) cells
 - Somatotrops → produce GH → generalized effects
 - Lactotrops → produce PRL → act on mammary glands
 - Gonadotrops → produce LH & FSH → act on ovaries & testes
 - Thyrotrops → produce TSH → act on thyroid gland
 - Corticotrops → produce ACTH → act on adrenal cortex
 - Secretion is regulated by hypothalamic hormones, as well as by feedback from the target glands

AP Hormones	Functions
GH	Promotes growth and development
PRL	Stimulates milk production
ACTH	Stimulates adrenal cortex to produce corticosteroids
TSH	Stimulates the thyroid gland to secrete T3/T4
LH	•Promotes ovulation + conversion of the ovulated ovarian follicle into corpus luteum (<i>produce progesterone until placenta takes over $\approx$ 2 weeks</i>) •Stimulates Leydig cells to secrete testosterone
FSH	•Stimulates growth and maturation of ova •Stimulates secretion of estrogen •Stimulates spermatogenesis

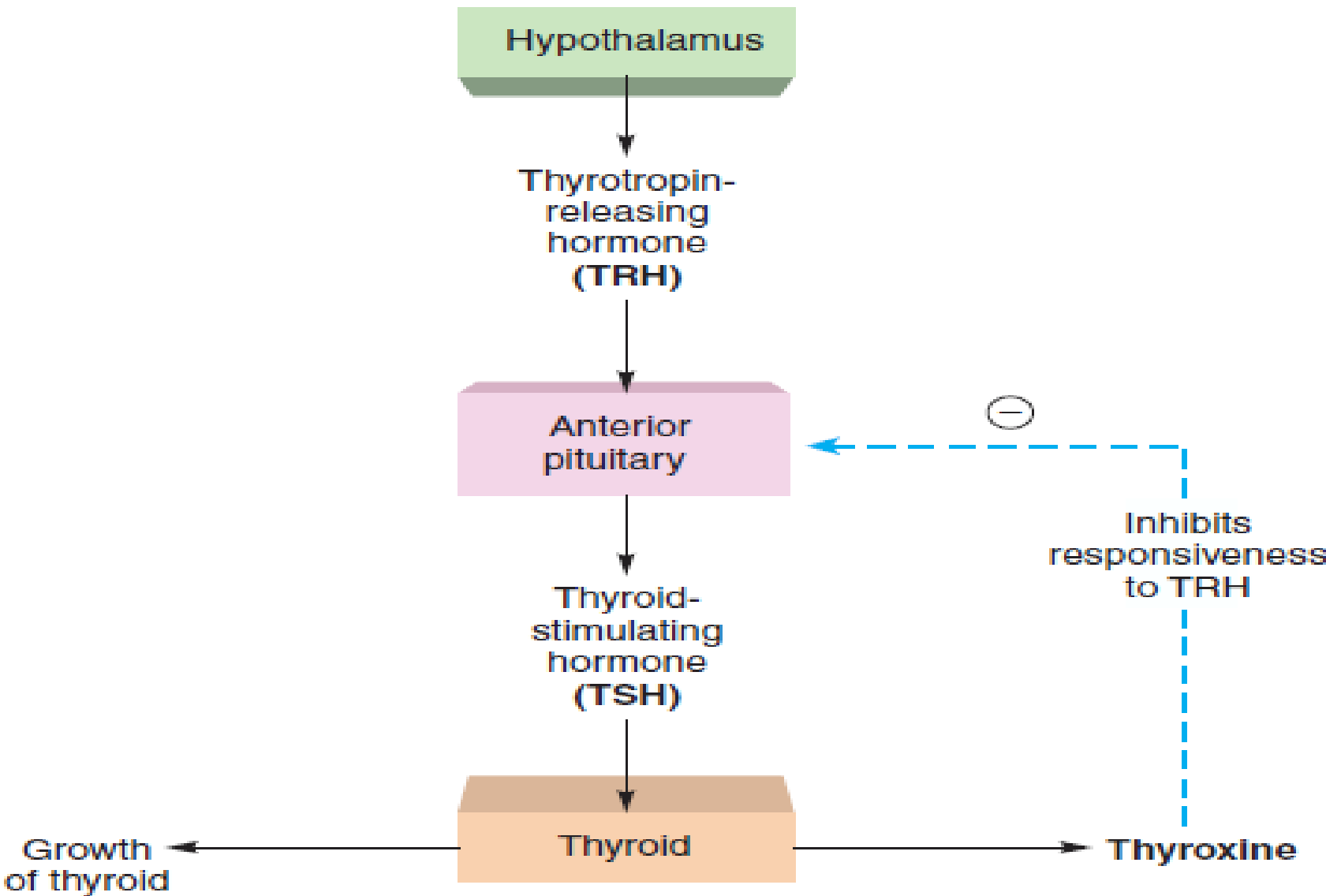


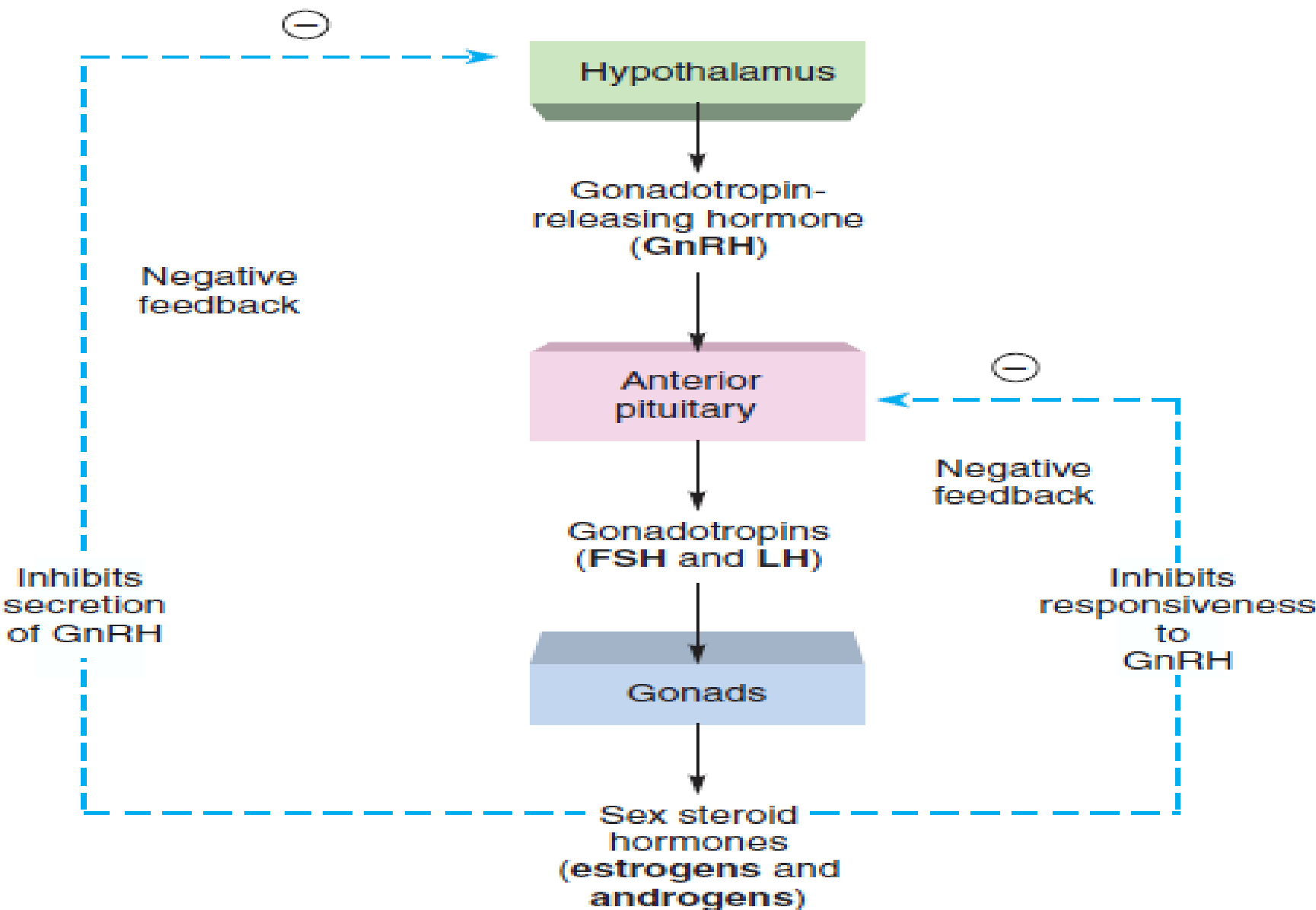
Pituitary Gland

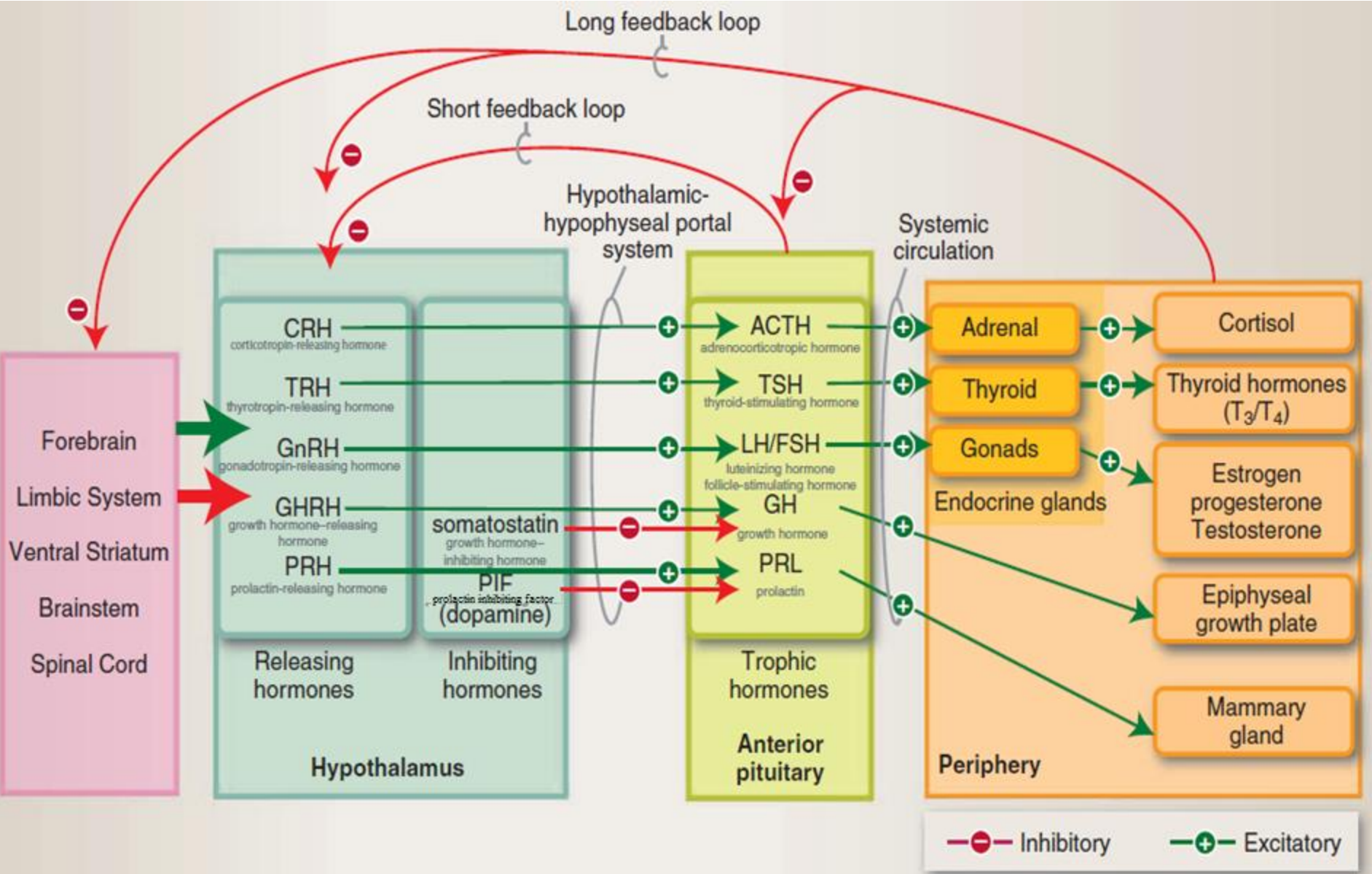
- Anterior Pituitary
 - Pituitary hormones have trophic effects:
 - High concentration of the anterior pituitary hormones causes target organ hypertrophy
 - Low levels cause target organ atrophy

Pituitary Gland

- **Feedback Regulation:**
 - Hypothalamus and anterior pituitary secretions are regulated by **-ve feedback** from the target organs
 - Under normal conditions, target glands exert an inhibitory effect on the hypothalamus/pituitary gland
 - Decreased secretion of target glands relieves the NFB inhibition on HT/APG → increase secretion of hypothalamic and pituitary hormones
 - Toward the middle of the menstrual cycle, the rising secretion of estradiol from the ovaries stimulates the anterior pituitary to secrete a “surge” of LH, which results in ovulation (**+ve feedback**)
 - Higher brain centers can affect the HT-APG-target organ axis







Pituitary Gland

- The “Dormitory Effect”
 - Tendency of synchronization of menstrual cycles of female roommates
 - Release of **Pheromones** from one person → sensed by Olfactory system of another person → modify reproductive cycle

Pituitary Gland

- **Posterior Pituitary**
 - An outgrowth of the hypothalamus composed of neural tissue
 - Hypothalamic magnocellular neurons (SON + PVN) synapse directly on to blood vessels in the posterior pituitary
 - They produce and secrete two peptide hormones directly into the posterior lobe capillaries
 - Paraventricular nuclei → **Oxytocin (OXT)**
 - Supraoptic Nuclei → **Vasopressin (ADH)**
 - Posterior pituitary stores and releases OXT and ADH

Pituitary Gland

- Posterior Pituitary
 - Stimulation of the hypothalamic nuclei initiate action potentials in their neurons → AP transmitted downward along the fibers from the SON/PVN on reaching the nerve ending → causes release of the hormone → hormones transported in combination with **neurophysins** (carrier proteins) → hormones absorbed into adjacent capillaries from which they reach the general circulation → *Neurosecretion*

Antidiuretic Hormone

- Arginine vasopressin (AVP)
 - Produced by the SON of the hypothalamus
 - Transported to the posterior pituitary for storage until release
 - Promote reabsorption of water in the renal tubules (DT & CT)
 - Causes vasoconstriction
 - Involved as a NT in fear response
- ADH release is stimulated by **Hyperosmolarity ($\downarrow$ ECV) + $\uparrow$ Ang-II...**

Antidiuretic Hormone

- ADH Disorders:
 - Diabetes Insipidus (DI):
 - Neurogenic DI → problem on hypothalamus/pituitary
 - Nephrogenic DI → problem on renal tubules
 - Manifestation → Polyuria and polydipsia

Oxytocin

- Produced by the paraventricular nuclei of the hypothalamus
- Secretion is stimulated by suckling reflex + coitus + uterine contraction
 - Stimulates contraction of the uterus during labor
 - Stimulates contraction of the mammary gland alveoli and ducts → milk-ejection in a lactating woman
 - Acts as a PRL releasing hormone
 - It helps sperm motility during ejaculation
 - It also acts as a NT in memory, fear conditioning

Growth Hormone

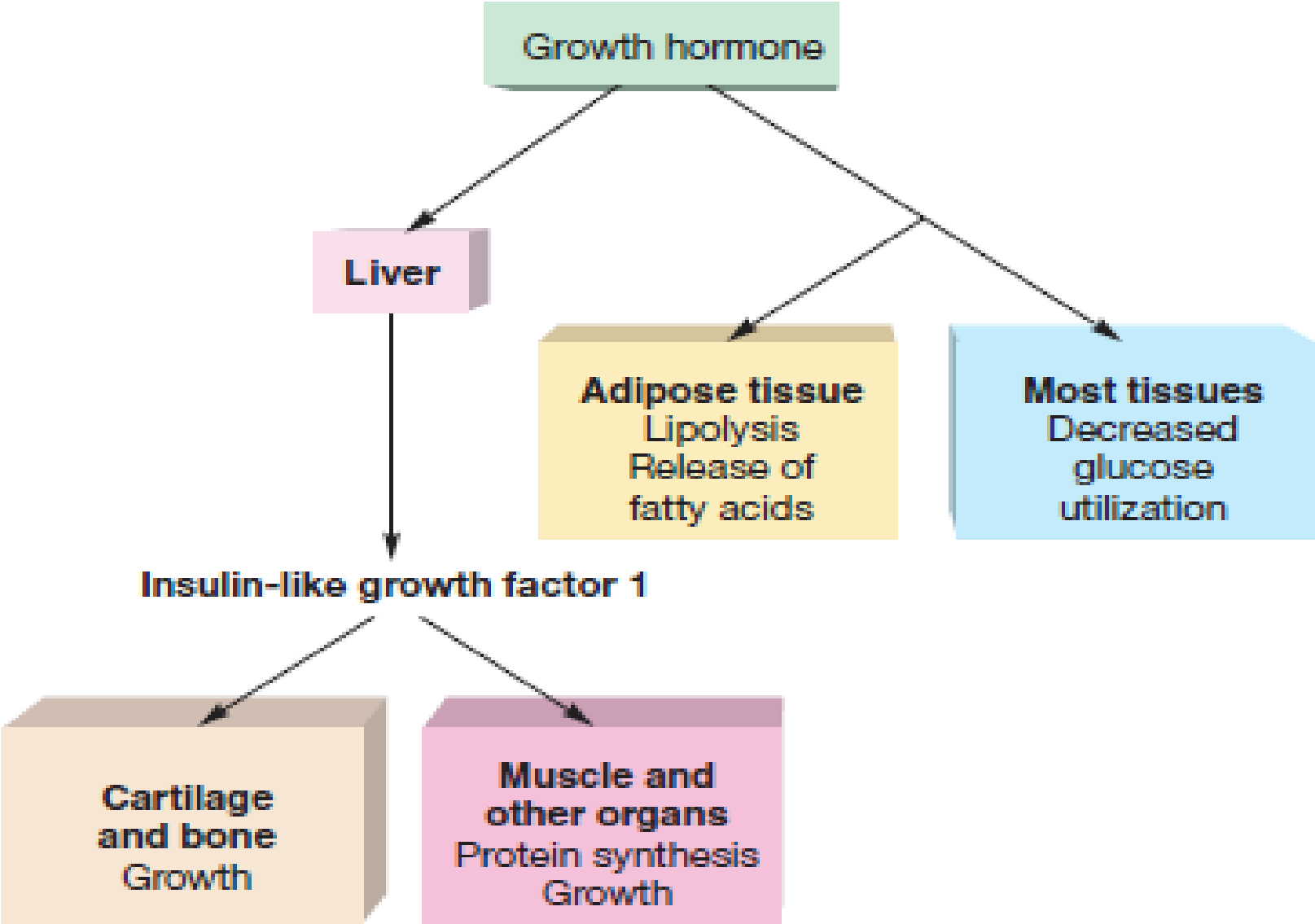
- Secreted in larger amounts by the APG than any other of its hormones
- **Factors stimulating GH secretion**
 - Growth hormone-releasing hormone from the hypothalamus
 - Decreased blood glucose and FFA
 - Increased amino acids
 - Fasting, starvation, protein deficiency
 - Trauma, stress, excitement, exercise
 - Hormones → testosterone, estrogen, PRL, glucagon, insulin, T3/T4
 - **Circadian variation** → ↑during sleep and ↓during wakefulness

Growth Hormone

- Factors inhibiting GH secretion
 - Somatostatin (GHIH)
 - Increased blood glucose and FFA
 - Obesity
 - Aging
 - Somatomedins (IGFs)

Growth Hormone

- Functions:
 1. Metabolic Regulation → Anabolic + Catabolic
 - a. Protein metabolism (anabolic)
 - ↓Protein catabolism
 - Promote the movement of aa into cells and incorporation of these aa into proteins (*like insulin*) → promote overall tissue and organ growth
 - In turn, it increases plasma concentration of glucose and FFAs to provide alternate energy source
- **NB:** The anabolic effect is particularly important during growing years → contributes to increase in bone length and tissue mass



Growth Hormone

b. CHO and Fat metabolism (catabolic)

- $\uparrow$ Glycogenolysis + $\uparrow$ Gluconeogenesis $\rightarrow$ $\uparrow$ Blood glucose level
- $\uparrow$ Lipolysis $\rightarrow$ $\uparrow$ FFA in plasma $\rightarrow$ $\uparrow$ Ketogenesis
 - Release of fatty acids from adipose tissue during periods of fasting (post-absorptive state, at night)
 - A rise in plasma FFA concentration results in decreased rates of glycolysis in many organs
 - Inhibition of glycolysis by fatty acids results in decreased glucose utilization by many tissues (+GH)

NB: GH acts to raise blood glucose concentration $\rightarrow$ “Diabetogenic” effect

Growth Hormone

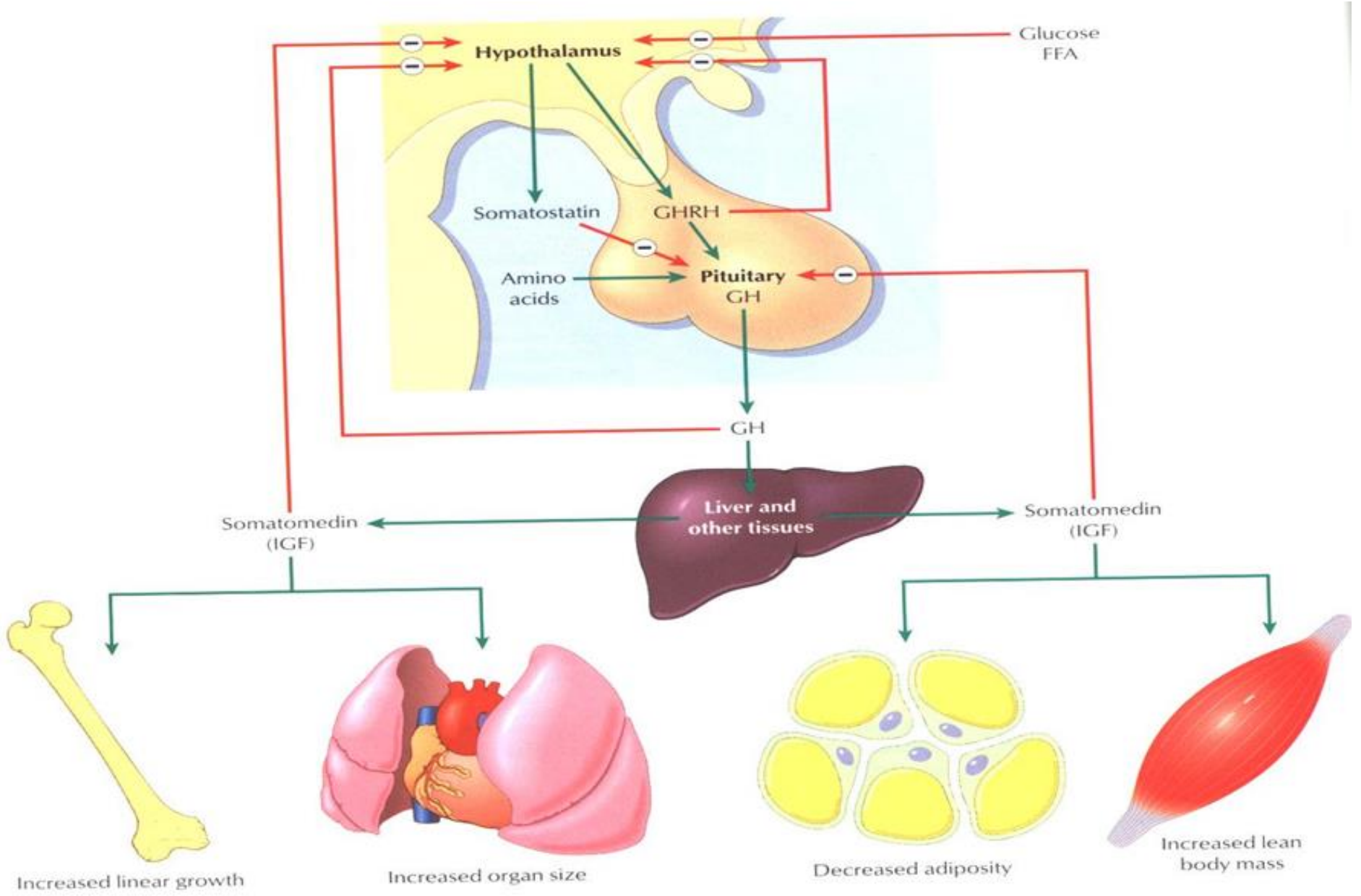
2. Growth Regulation

- Stimulate bone growth
- Stimulate mitosis in the epiphyseal discs of cartilage
- Mediated by Somatomedins (IGF-1 + IGF-2) → stimulate the chondrocytes to divide and secrete more cartilage matrix
 - Part of this growing cartilage is converted to bone, enabling the bone to grow in length
- IGFs have insulin-like effects and serve as mediators for some of growth hormone's actions

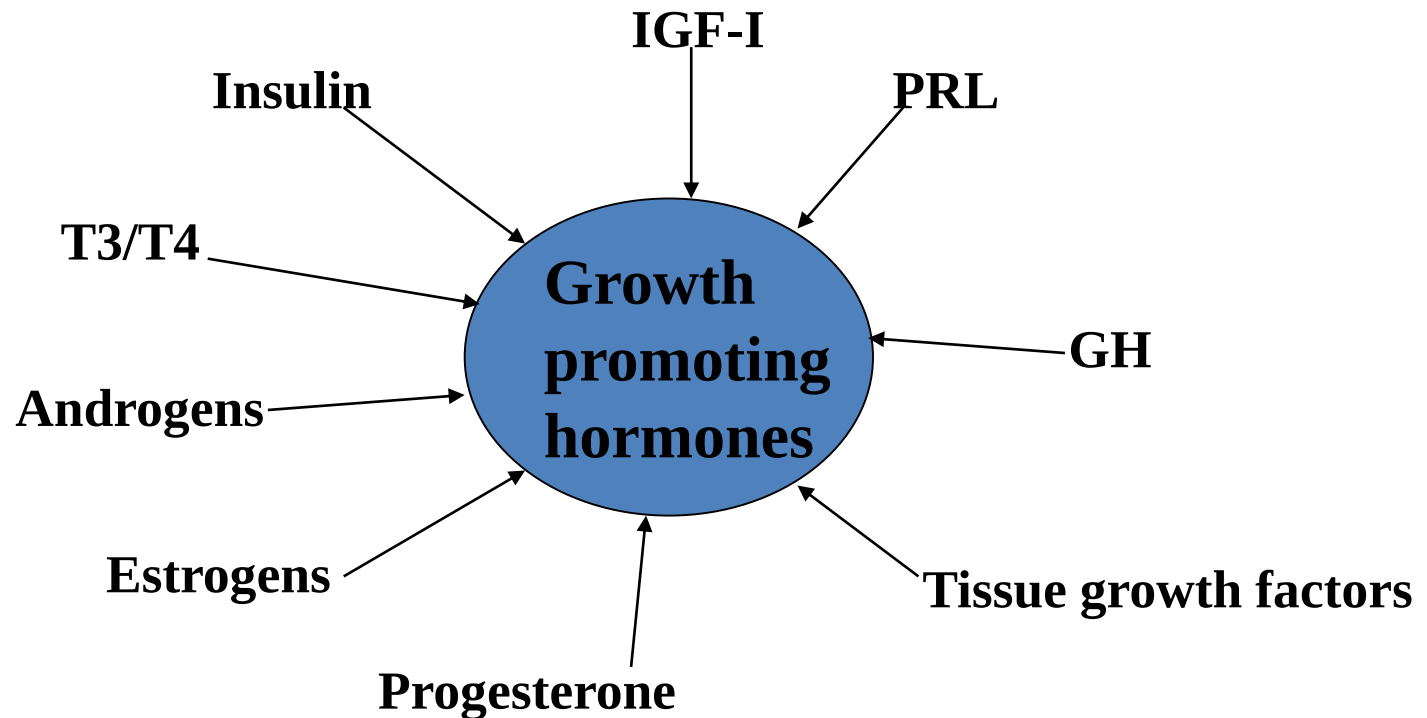
Growth Hormone

2. Growth Regulation

- **IGF-1** is produced by liver in response to GH stimulation
 - Stimulates cell division and growth
- Chondrocytes (cartilage cells) also produce IGF-1 in response to GH stimulation → functions as an **autocrine regulator**
- IGF-2 has more insulin-like actions



Factors affecting growth



- Other factors affecting growth
 - Hereditary (genetic factors) + Nutritional factors

Growth Hormone

3. Other roles:

- $\uparrow$ RBC production
- $\uparrow$ MR $\rightarrow$ Thermogenic
- $\uparrow$ Insulin secretion
- $\uparrow$ Secretion of Somatomedin-C (IGF)
- Electrolytes $\rightarrow$ $\uparrow$ Absorption of Na^+ , K^+ and Ca^{2+} ...

Summary

Hormone	Target Tissue	Principal Actions	Regulation of Secretion
ACTH (adrenocorticotrophic hormone)	Adrenal cortex	Stimulates secretion of glucocorticoids	Stimulated by CRH (corticotropin-releasing hormone); inhibited by glucocorticoids
TSH (thyroid-stimulating hormone)	Thyroid gland	Stimulates secretion of thyroid hormones	Stimulated by TRH (thyrotropin-releasing hormone); inhibited by thyroid hormones
GH (growth hormone)	Most tissue	Promotes protein synthesis and growth; lipolysis and increased blood glucose	Inhibited by somatostatin; stimulated by growth hormone–releasing hormone
FSH (follicle-stimulating hormone)	Gonads	Promotes gamete production and stimulates estrogen production in females	Stimulated by GnRH (gonadotropin-releasing hormone); inhibited by sex steroids and inhibin
PRL (prolactin)	Mammary glands and other sex accessory organs	Promotes milk production in lactating females; additional actions in other organs	Inhibited by PIH (prolactin-inhibiting hormone)
LH (luteinizing hormone)	Gonads	Stimulates sex hormone secretion; ovulation and corpus luteum formation in females; stimulates testosterone secretion in males	Stimulated by GnRH; inhibited by sex steroids